



Integrating climatic response in competition dependent tree-level growth models for northern hardwoods



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ABSTRACT

With increased rates of climate change, it is imperative for forest managers to have access to models that can take into account the expected effects of climate change on tree growth. To this end, growth function are sometimes used that include climatic variables such as mean annual temperature or precipitation averaged over decades. Such growth models are usually relatively easy to develop but they do not take into account the fact that tree diameter growth on a given year is determined not by climatic conditions that prevailed up to 30 years before but mainly by climatic conditions that prevailed during the current and previous year. Our objective is determine if including climatic variables obtained from dendroclimatic response function will lead to growth models having a better fit to data than versions with 30-year average climatic conditions, or no climate at all. Growth models were developed for *Betula alleghaniensis*, *Acer saccharum*, *Acer rubrum* and *Fagus grandifolia* using data from south-eastern Quebec. Three types of growth function were compared. A first set of growth function was developed in which the potential growth of a tree was modeled as a function of tree size and site characteristics (vegetation type and drainage) to be further modified as a non-linear function of plot basal area. The effect of climate was not explicitly accounted for in this first set of growth function, therefore they will be referred to as Climate-implicit models. A second set of growth function was developed in which we explicitly accounted for the effect of climate by incorporating 30-year mean annual temperature and precipitation in the growth function. In a third type of growth function, also climate-explicit, we incorporated the most significant recent climatic variables identified using climatic response function developed for each species based on dendrochronological and climatic data. The three types of models were compared based on the Akaike information criterion (AIC). Our results showed that Climate-explicit growth models with climatic variables obtained from response function analysis outperformed other growth models for three out of four species (*B. alleghaniensis*, *A. saccharum* and *F. grandifolia*). Incorporating climate in the form of 30-year average climatic conditions brought some improvement over a non-climatic function for *A. rubrum*, but this was not the case for other species. Accounting for growth dependency on climate by including recent monthly climatic variables provided by response function could be a potentially useful approach for the development of a new lineage of tree growth models dealing with climate change.

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1. Introduction

Given the current and anticipated changes in climatic conditions, tree growth models need to account for the influence of climate on growth, especially if models are intended to be used

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to predict growth over several decades. Several tree growth models do account for the effect of climate on growth. Moreover, growth models which explicitly incorporate climate effects on growth generally show a better fit to data than models without climate terms. For instance, [Huang et al. \(2013\)](#) showed that the inclusion of a 30-years climate variable significantly improved model fit for two deciduous species (*Populus tremuloides* Michx. and *Betula papyrifera* Marshall.) in eastern Canada. Mixed models using climatic input were computed by [Laubhann et al. \(2008\)](#) and [Rayamajhi and Kush \(2006\)](#) for deciduous species in Europe and

USA. They found that including climatic variables improved growth model predictions compared to growth models that did not account explicitly for climate effects. However, in all of these studies, climate was represented by a 20–30 year average of annual or seasonal climatic data.

Assessment of climate change impacts on predicted growth trends need to be derived from models which explicitly incorporate the effects of climate on growth. Dendrochronology, which deals with long-term records of tree growth under natural environmental conditions, can be used to evaluate the impacts of climatic change on forest productivity (Schweingruber, 1996). Dendroclimatic analysis (e.g. response function) can be useful as an exploratory tool for determining which climatic variables are relevant to use and for identifying periods (months and/or seasons) when climatic conditions (temperature and/or precipitation) are more likely to influence tree growth (Schweingruber, 1996; Hofgaard et al., 1999; Tardif et al., 2001; Deslauriers et al., 2003; Girard et al., 2011a). For example, dendroclimatic response function have been found to give good results in modeling past and future radial growth (Tardif et al., 2001).

According to International Panel on Climatic Change workgroup (IPCC) scenarios, southern Québec will experience increased rates of climate change for the 21st century (Solomon et al., 2007). Not only are changes expected in terms of mean annual temperatures, but also in terms of monthly variations and frequency of extreme climatic events. This is crucial to consider since diameter growth of most North American trees is mainly influenced by prevailing climatic conditions during a few months of the previous and current year, because of pre- and neoformation of cambial cells (Schweingruber, 1996; Hofgaard et al., 1999; Tardif et al., 2001; Deslauriers et al., 2003; Girard et al., 2011a). Therefore, including climatic variables such as mean annual temperature and precipitation in tree growth models, especially when the value of these variables is obtained from averaging several decades of data, may not capture the effects that expected changes in monthly temperatures and precipitation may have on tree ring growth. Using climatic response function (Schweingruber, 1996) to identify the monthly climatic variables most important to a species diameter growth, and including these combinations of monthly climatic variables in tree growth models might prove to be a promising approach to account for climatic change in competition-dependent tree growth models, for example the USDA's Forest Vegetation Simulator or 3-PG (Landsberg and Waring, 1997).

The objective of this study is to test the hypothesis that including climatic variables obtained from dendroclimatic response function will lead to tree growth models having a better fit to data than model versions in which climate is only represented by 30-year average conditions, or not represented at all. To address our objective, tree growth models were developed for four deciduous tree species found in northern hardwood forests in southern Québec: *Betula alleghaniensis* Britton (yellow birch), *Acer saccharum* Marsh. (sugar maple), *Acer rubrum* L. (red maple) and *Fagus grandifolia* Ehrh. (American beech). Growth models were calibrated using forest inventory data as well as dendrochronological material. Model development was performed in three steps. First, Climate-implicit models (i.e., models in which climate is not explicitly accounted for) were calibrated for each species with plot-level basal area (used as the competition variable), tree size and site conditions (vegetation type and drainage). Second, dendroclimatic response functions were established for each species from an analysis of tree cores taken on a subset of trees. The response functions were used to identify the suites of climatic variables (temperature and precipitation of specific months of the current and/or previous growing season) that most significantly influenced annual ring formation for each species. Third, Climate-explicit versions of growth models were developed by incorporating the suites of climatic variables

identified in the response function. An alternative approach to include climate in growth models was also considered in which 30-year mean annual climate variables (temperature and precipitation) were used.

2. Methods

2.1. Study area

The study area is located in southeastern Québec and is part of the sugar maple-basswood bioclimatic domain (Robitaille and Saucier, 1998) (Fig. 1). The study area extends from 72°W 45.25°N to 70.3°W 46°N, for a total of 10845 km². Mean elevation of the plots is 431 m above sea level (range: 214–639 m). For the period 1980–2010, climatic conditions were characterized by a mean annual temperature of 4.7 °C and annual precipitation averaging 1335 mm, of which 20% falls as snow. The growing season length varies between 180 and 190 days (Environnement Canada, 2012).

2.2. Database

We used a forest inventory database provided by Domtar Inc., which included data from 4592 permanent plots sampled at different times between 1963 and 2010. Most of the plots were circular (radius of 11.28 m) but some were square (20 m × 20 m), and they all covered an area of 400 m². We selected plots that met the following criteria: (1) plots located in landscape units 8, 30, 31 or 32 (Robitaille and Saucier, 1998); (2) plots with a potential vegetation type that was either deciduous or mixed (based on forest maps); (3) plots which comprised >75% of total basal area as deciduous trees; (4) plots which were sampled at an interval of ≥5 years; and (5) plots with drainage ranging from poor to moderate.

A total of 590 plots met our criteria, corresponding to 15742 tree measurements. The main tree species found in the selected plots were: *B. alleghaniensis* (3346 measurements on 2213 trees), *A. rubrum* (3180 measurements on 2315 trees), *A. saccharum* (8368 measurements on 5278 trees) and *F. grandifolia* (848 measurements on 504 trees). In the dataset, mean diameter at breast height (dbh) was 187 mm (91–735 mm) for *B. alleghaniensis*, 208 mm (91–622 mm) for *A. rubrum*, 201 mm (91–704 mm) for *A. saccharum* and 185 mm (91–515 mm) for *F. grandifolia*. Plot-level basal area (BA) ranged from 8 to 58 m² ha⁻¹. Soils were typically brunisols (occasionally podzols) covered by a Moder humus layer.

For each dbh measurement i recorded in the database, we estimated the proportion of the growing season (P_{season_i}) elapsed at the sampling date ($date_i$) using the logistic function presented in Duchesne and Ouimet (2008):

$$P_{season_i} = \frac{1}{(1 + \exp^{(12.54 - 0.066 \cdot date_i)})} \quad (1)$$

where P_{season_i} ranges from 0 to 1 and where $date_i$ is expressed in Julian days.

The interval between two measurements was then calculated as:

$$Interval_{0-1} = (Year_1 - Year_0) + (1 - P_{season_0}) - (1 - P_{season_1}) \quad (2)$$

where Interval is expressed in number of growing seasons, $Year_i$ is the measurement calendar year and P_{season_i} is from Eq. (1).

Annual diameter growth (mm year⁻¹) of each tree between two measurements was calculated by dividing the total increase in dbh ($dbh_1 - dbh_0$) by the interval length between measurements ($Interval_{0-1}$). Occasionally, negative dbh increments were

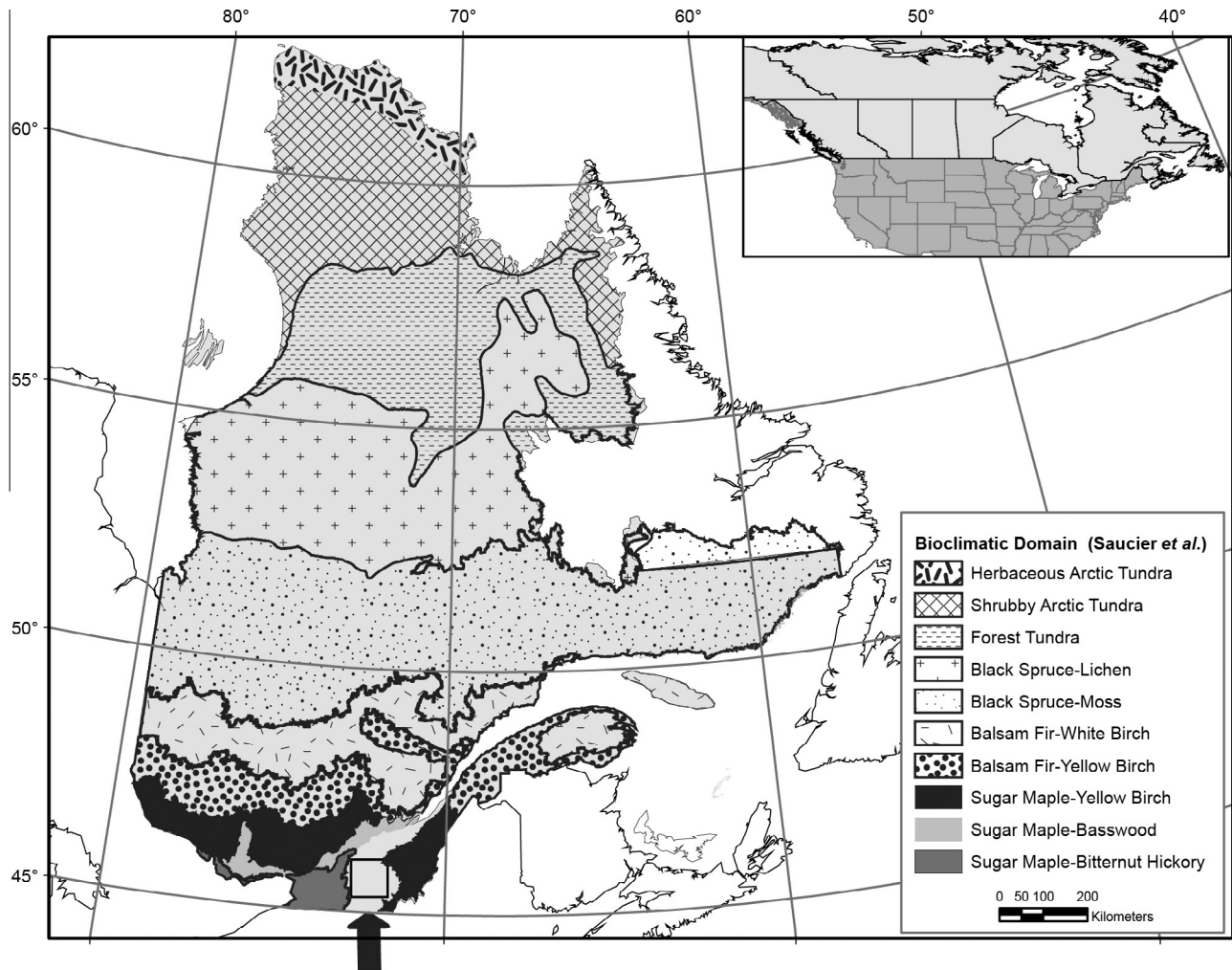


Fig. 1. Location of the study area in southern Québec (Canada) was included in the box highlighted by the black arrow: from 72°W 45.25°N to 70.3°W 46°N. Bioclimatic domains are shown according to [Saucier et al. \(1998\)](#).

obtained, possibly due to field measurement errors. For statistical analysis, only positive dbh increment values ($\geq 0 \text{ mm year}^{-1}$) were used, which meant that 369 negative values were removed from the dataset. Since excluding negative growth values from analysis could lead to a bias in growth estimation (leading to an overestimation of mean growth), we also removed the highest 369 growth values from the dataset, thereby performing a symmetrical truncation of the distribution. Remaining dbh increments ranged from 0 to $9.79 \text{ mm year}^{-1}$.

Basal area ($\text{m}^2 \text{ ha}^{-1}$) was calculated for each plot, at each measurement date, and included only living trees with dbh $> 90 \text{ mm}$. Two different binary variables were derived from the ecological type ([Gosselin, 2007](#)) assigned to each plot. First, a drainage effect (DE) variable was defined where $\text{DE} = 0$ represented subhydryc plots (corresponding to drainage classes 4 and 5), whereas $\text{DE} = 1$ represented mesic plots (corresponding to drainage classes 1, 2 and 3). Second, a vegetation type effect (VE) variable was assigned to each plot where $\text{VE} = 0$ represented mixed forests and where $\text{VE} = 1$ represented deciduous forests.

2.3. Dendrochronological sampling and response function

To develop species-specific climatic response function, we had to be able to relate annual ring growth to climatic conditions that prevailed during the current and previous growth seasons. The forest inventory database described above, based on plot

re-measurements performed at intervals ≥ 5 years, could not be used to this end; tree core data were required. To obtain such data, five inventory plots were selected from across the study area (one was at the geographic center and one each at the easternmost, westernmost, northernmost and southernmost parts of the study area). In each of these five plots, the ten largest trees of each of the four study species were sampled at dbh level (1.3 m above-ground) using a Pressler corer, for a total of 50 tree cores per species. All cores were glued to a wooden support and finely sanded to make the xylem cells clearly visible under a 40X binocular microscope. Growth rings were counted on each core and diagnostic rings (light rings, frost rings, incomplete rings, etc.) were identified to facilitate dating. Ring widths ($\pm 0.002 \text{ mm}$) were measured using the WinDendro software (Régent Instruments Inc., Québec, Canada). Ring-width series were cross-dated visually and a second check was made using the statistical software COFECHA ([Holmes et al., 1986](#)). Each ring-width series was standardized using the software package ARSTAN ([Cook, 1985](#)). These procedures were used to eliminate non-climatic variation. A double detrending was performed using a negative exponential or a straight line, as well as a cubic spline function with a 50% cutoff wavelength. The standardization process allows age-related tendencies to be removed from the growth curve to obtain a homogenized variance. The resulting data consists of a dimensionless, standardized index with variations representing the effect of stress factors (climate, insect herbivory, etc.) on tree growth. Once ring-width series were

accurately cross-dated and standardized, the series were averaged for each plot and master chronologies were constructed for each species by averaging the index series of all trees (Holmes et al., 1986). Master chronologies were calculated for each plot using cores that were easily cross-dated with other plots (correlation between dendro-series >0.65 instead of >0.5 typically used in literature (Schweingruber, 1996)). Residual chronologies were used instead of indexed series to remove most of the auto-correlation in the dataset (Schweingruber, 1996).

Relationships between residual dendro-series and climate (Fig. 2) were investigated using partial least square (PLS) regression (Wold, 1995; Amato and Vinzi, 2003). PLS regression was chosen because it handles many variables with relatively few observations (Cramer et al., 1988) and deals with correlated variables (Wold 1995). The use of residual series from the standardization copes with autocorrelation. The number of components was chosen using a permutation test (Good, 1994) with a 5% threshold for the explained variance. Variables were tested with a 50000-step cross-validation test (Amato and Vinzi 2003). Variables were retained only when the 95% confidence interval for their partial correlation coefficient excluded zero. Climatic monthly variables tested for the regression were precipitation (P) and mean temperatures (MT) from June of the previous year ($n-1$) to September of the current year (n), and this was performed for the 1980–2011 period. Climatic data for the 1980–2011 period were obtained for each inventory plot (and hence for each tree) using the BioSim 10 model (Régnière and St-Amant, 2007; Solomon et al., 2007). In BioSim 10, climatic data are interpolated from the five nearest weather stations. The low number of observations (30 years of climate and growth data) and high inter-annual variability of climate made grouping monthly climatic variables necessary to obtain significant correlations. We grouped successive months having the same sign for their individual partial correlation coefficients. Statistical analyses were performed with ADE4 package in R software (Dray and Dufour, 2007; R Development Core Team, 2011).

2.4. Three different types of growth models

Diameter growth models were developed where growth was modeled as an additive or multiplicative combination of potential growth, and tree size and competition effects (Weiskittel et al.,

2011). The potential growth itself was modeled as an additive combination of an intercept (b_0), a vegetation type effect (VE) and a drainage effect (DE) (Eq. (3)). The symbol ε represent the error term of the equation:

$$\text{Potential growth} = b_0 + b_1 VE + b_2 DE + \varepsilon \quad (3)$$

Both additive (Eqs. (4) and (5)) and multiplicative (Eq. (6)) growth models were tested where potential growth was modulated by tree size (size effect, SE) and competition (basal area effect, BAE). The choice of testing both model was simple: additive and multiplicative forms are widespread to model Canadian forest growth. The demonstration in this study had to be exhaustive in order to cover every possibilities. The symbol ε represent the error term of the equation:

$$\text{Diameter growth}_i = \text{Potential growth} + SE + BAE + \varepsilon \quad (4)$$

$$\text{Diameter growth}_j = (\text{Potential growth} + SE) \cdot BAE \cdot \varepsilon \quad (5)$$

$$\text{Diameter growth}_k = \text{Potential growth} \cdot SE \cdot BAE \cdot \varepsilon \quad (6)$$

For additive models, the linear and quadratic equation forms of size and basal area effects were tested (Table 1). Multiplicative models were also computed using a logistic form for the size effect, and three multiplicative equation forms for the basal area affect (exponential, logistic and lognormal). Mathematical forms used for the growth models were adapted from previous studies (Pacala et al., 1994; Canham et al., 2004, 2006; Fortin et al., 2008; Fortin et al., 2009; Fortin and Langevin, 2010). All possible combinations of terms were tested (including the possibility that a variable had no influence on tree growth) (Fig. 3). The resulting set of growth models ($n = 80$ models) will be referred to as “Climate-implicit models”.

Next, the potential growth term of each of the 80 models was modified to include a climatic component. First, 30-year mean annual temperature (i.e. Temperature effect, TE) and precipitation (i.e. Precipitation effect, PE) were used to modify in a linear manner the potential growth term (Eq. (7)). The symbol ε represent the error term of the equation:

$$\text{Potential growth} = b_0 + VE + DE + PE + TE + \varepsilon \quad (7)$$

As trees were sampled at different times during the 1963–2010 period, mean annual temperature and total precipitation were calculated for their respective 30-year period. For example, if the last sampling was done in 1993 for a particular tree, the 1963–1993 time period was used. The 80 resulting growth models will be referred to as “Climate-explicit-30-years models”.

Alternatively, for each species, variables identified as significant by the PLS response function were included in the “potential growth” term (Eq. (8)). The symbol ε represent the error term of the equation:

$$\text{Potential growth} = b_0 + VE + DE + PE_{RF_i} + TE_{RF_i} + \varepsilon \quad (8)$$

The climatic variables were added as a linear function to preserve the equation form of the response function. Response function climate variables (RF variables hereafter) added in the growth models were the following, for each species:

$$PE_{RF_{B.alleganiensis}} = (c_1 \cdot PE_{-6,-7,-8}) + (c_2 \cdot PE_{-9,-10}) + (c_3 \cdot PE_{2,3}) + (c_4 \cdot PE_{8,9}) + \varepsilon \quad (9a)$$

$$TE_{RF_{B.alleganiensis}} = (c_5 \cdot TE_{-7,-8,-9}) + (c_6 \cdot TE_{7,8}) + \varepsilon \quad (9b)$$

$$PE_{RF_{A.rubrum}} = (c_7 \cdot PE_{1,2,3}) + (c_8 \cdot PE_8) + \varepsilon \quad (9c)$$

$$TE_{RF_{A.rubrum}} = (c_9 \cdot TE_{-8}) + \varepsilon \quad (9d)$$

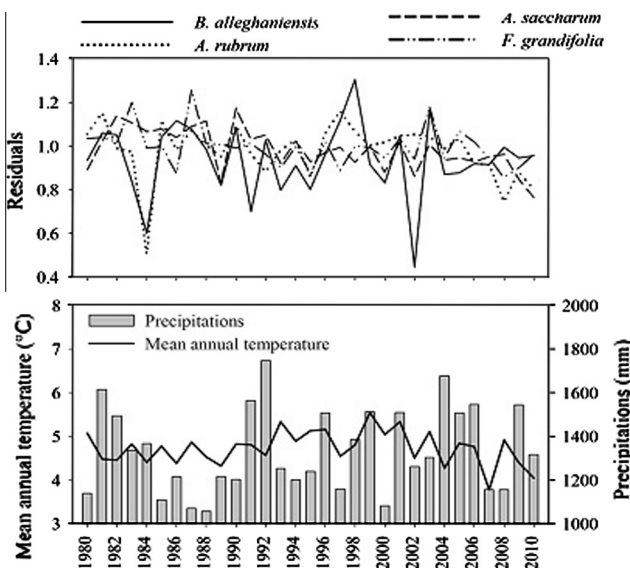
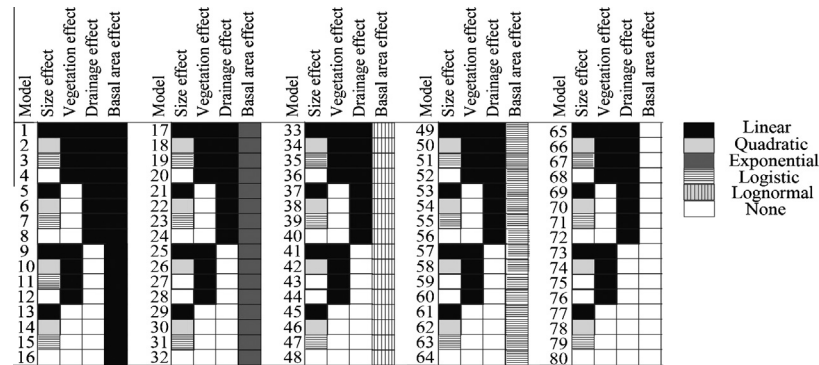


Fig. 2. Up: Residual annual growth index for each species. Down: Annual monthly precipitation and mean monthly temperature for the response function calibration period (1980–2011).

Table 1

Description of the various mathematical forms that were tested for each type of effect (size, vegetation type, drainage and basal area).

Mathematical form	Size effect	Vegetation effect	Drainage effect	Basal area effect
Linear	$c \cdot DBH$	$b2 \cdot VE$	$b3 \cdot VE$	$b1 \cdot BA$
Quadratic	$(c \cdot DBH) + (d \cdot DBH^2)$	–	–	–
Logistic	$\frac{1}{\left(1 + \left(\frac{DBH}{x_0}\right)^{-\eta}\right)}$	–	–	$\frac{1}{\left(1 + \left(\frac{BA}{x_c}\right)^{-\eta}\right)}$
Exponential	–	–	–	$\exp(-f \cdot BA^h)$
Lognormal	–	–	–	$\exp\left(-0.5 \cdot \left(\log_{10}\left(\frac{BA}{x_c}\right)\right)^2\right)$

**Fig. 3.** Graphical representation of the 80 combinations of effects tested and their various mathematical forms. Black squares represent a linear effect, light gray squares represent a quadratic effect, dark gray squares represent an exponential effect, horizontal lines a logistic effect and vertical lines a lognormal effect. The white squares represent the possibility of a null effect on tree growth.

$$PE_{RF_{A.saccharum}} = (c_{10} \cdot PE_{-6}) + \varepsilon \quad (9e)$$

$$TE_{RF_{A.saccharum}} = (c_{11} \cdot TE_2) + (c_{12} \cdot TE_4) + \varepsilon \quad (9f)$$

$$PE_{RF_{F.grandifolia}} = (c_{13} \cdot PE_{-7,-8}) + \varepsilon \quad (9g)$$

$$TE_{RF_{F.grandifolia}} = (c_{14} \cdot TE_{1,2}) + (c_{15} \cdot TE_4) + (c_{16} \cdot TE_{6,7}) + (c_{17} \cdot TE_{-8}) + \varepsilon \quad (9h)$$

where PE and TE represent precipitation and temperature effect respectively. Subscripts following PE and TE terms represent month of the previous year (minus sign) or current year, and c_i represent parameters to be estimated. The symbol ε represent the error term of the equation. The 80 resulting growth models were called “Climate-explicit-RF models”, where “RF” stands for “Response function”.

A total of 240 models were computed per species (80 Climate-implicit, 80 Climate-explicit-30-years, and 80 Climate-explicit-RF models), resulting from combinations of three equation forms for the size effect, four equation forms for the basal area effect (competition effect), and linear effects of vegetation and drainage. Using 75% of the dataset (data randomly chosen, 25% was saved for validation), each of the 240 growth models was fit to data and models were compared in an AIC analysis (Akaike, 1974) using the NLMIXED procedure in SAS 9.3 software (SAS Institute, 2012). According to Burnham and Anderson (2002, 2004), it would be possible to compare up to a few thousands different models using AIC. Models were ranked by AIC_c (AIC_c is AIC with a correction for finite sample sizes) value and the model with the lowest AIC_c value was considered the one that provided the best fit to data. Delta AIC_c values (ΔAIC_c) were then calculated between each model's AIC_c and the AIC_c of the best model, for a species. Models that did not differ by more than two units of AIC_c can be considered as providing equivalent evidence (Burnham and Anderson, 2002).

Akaike weights and evidence ratios were also computed. The Akaike weight (Eq. (10)) represents the relative likelihood of a model and the strength of evidence for each model. Akaike weights

are also useful to give a measure of the relative importance of a variable. The interpretation of Akaike weights is straightforward: they represent the ratio of the ΔAIC_c of a given model relative to the whole set of R candidate models.

$$Akaike_weight = \omega_i = \frac{\exp(-\Delta_i/2)}{\sum_{r=1}^R \exp(-\Delta_r/2)} \quad (10)$$

One can also compare the Akaike weights of the « best » model and competing models to determine to what extent one is better than another. These are termed evidence ratios and are calculated as:

$$Evidence\ ratio = \frac{\omega_j}{\omega_i} \quad (11)$$

where model j is compared against model i . For example, an evidence ratio of 1.375 would indicate that model j is only 1.375 more likely than model i to be the best, given the set of candidate models and the data (Burnham and Anderson, 2004; Mazerolle, 2006). In other words, there would be a high degree of uncertainty regarding the best model.

Finally, using the 25% saved part of the dataset, a cross validation was made to evaluate bias (Eq. (12)) and model accuracy (Eq. (13)) (Walther and Moore, 2005).

$$Bias = \sum (Observed - Predicted)/n \quad (12)$$

$$Accuracy = \sum ((Observed - Predicted)^2)/n \quad (13)$$

3. Results

3.1. Response function

In our study area, the correlation among individual dendro-series used to calibrate the response function was >0.65 (for 50 trees per species, results not shown), indicating that trees from different parts of the study area were influenced by the same climatic

conditions (Schweingruber, 1996). Therefore, grouping dendro-series from the five different plots (in which tree cores were sampled) allowed us to establish valid response function for the whole study area. This generalization of the calculation eliminates the necessity to calibrate growth models with a longitude or a latitude term. Moreover, the climate of the last 30 years showed no significant annual or monthly trend over time in mean annual temperatures and precipitation (results not shown) indicating that the response function calibration period is adequate. In this area, climate change is observed by an increase of extreme climatic events rather than an annual trend. The relatively high correlation among dendro-series suggests that both response function and growth models are valid for the whole study area.

Relatively strong associations were observed between tree-ring growth and climatic monthly data, for all four species (Fig. 4). Overall, the significant climatic variables entered in the response function models accounted for 51%, 48%, 42% and 40% of the variation in annual ring growth for *B. alleghaniensis*, *A. rubrum*, *A. saccharum*, and *F. grandifolia*, respectively. For *B. alleghaniensis*, precipitation in June, July and August of the previous year were negatively associated with ring-width. Significant associations were found with the combined precipitation of September and October of the previous year, with the combined precipitation of February and March of the current year, as well as with mean temperatures in July, August and September of the previous year. Precipitation in August and September as well as mean temperatures in July and August of the current year were positively associated with ring-width. Three climatic variables were found significant for *A. rubrum*: a negative association with precipitation in January, February and March of the current year, and positive associations with precipitation and mean temperature during August of the current year. For *A. saccharum*, precipitation in June of the previous year, and mean temperature in February and April of the current year were positively linked with ring-width. Finally, for *F. grandifolia*, precipitation in July and August of the previous year were positively related to ring-width. Positive relationships were found for mean temperatures in January, February, April and August of the current year, while a negative link was found with summer temperatures in June and July of the current year. Significant variables of each response function described above were entered in Climate-explicit-RF growth models to modify the potential growth.

3.2. Comparison of Climate-implicit, Climate-explicit-30-years and Climate-explicit-RF growth models

Among the 240 different combinations of models, the models that were retained as providing the best fit to data (with lowest AIC_c , as well as with $\Delta AIC_c < 2$) were Climate-explicit-RF models for three of our four study species, i.e., *B. alleghaniensis*,

A. saccharum and *F. grandifolia* (Table 2). The trend was also clearly observable when the sums of Akaike weights were calculated by model type and compared (Table 3). Such results indicated that for these three species, Climate-explicit-RF models outperformed both Climate-implicit and Climate-explicit-30-years models. This result is also supported by the fact that second models ($AIC_c < 10$) were Climate-explicit-RF for *A. saccharum* and *F. grandifolia*. For *B. alleghaniensis*, second models were Climate-explicit-30-years models but with an $AIC_c > 6$ (results not shown).

However, integration of climatic variables did not improve performance of growth models for *A. rubrum*. For this species, a much larger number of models could be considered as equivalent (i.e., with $\Delta AIC_c < 2$), and those models comprised both Climate-implicit, Climate-explicit-30-years and Climate-explicit-RF models. In fact, among the three types of models, the Climate-implicit ones were more frequently retained. Climate-explicit-30-years models also showed some support.

Inclusion of size effect improved model fit for all species. In fact, all models without this variable had lower Akaike weights. Potential vegetation (VE variable) was never of a great support in most models. However, lower Akaike weights were observed when VE was removed from *F. grandifolia* growth models. Such a result is probably simply due to the rare occurrence ($n = 2$) of this species in mixed stands ($VE = 0$). Inclusion of the drainage effect (DE) variable was important for *B. alleghaniensis* and *A. saccharum* but not for the two other species. Finally, including a competition effect (here, plot basal area) in the growth models clearly improved model fit since for all four species, models with a competition term had higher Akaike weights. For each of the four species, the best model that did not included the competition term differed by more than 25 units of AIC_c (ΔAIC_c) from the best model overall (result not shown).

For *B. alleghaniensis*, among the 240 models tested, the best fit ($\Delta AIC_c < 2$) to the data was observed for two models (23 and 19) (Table 2). Each of these models included a logistic size effect, an exponential basal area effect, and a drainage effect (Table 2). Model 23 was 1.871 more likely than the model 19, and did not comprise a vegetation type effect while the latter did.

Model comparison for *A. rubrum* showed that nine Climate-implicit, eight Climate-explicit-30-year models and one Climate-explicit-RF provided equivalent evidence (i.e., with a $\Delta AIC_c < 2$) (Table 2). Overall, size effect was most often retained in a logistic or quadratic form. Basal area was always included in the best models, but never in a linear form. Drainage and vegetation were included in only three of the eighteen “best” models (Table 2). For *A. rubrum*, addition of climatic variables such as average 30-year precipitation and temperatures did not significantly improve model performance (see models 30, 31, 46, 47, 62 and 63) as Climate-implicit and Climate-explicit-30-years model had equivalent support in the data. Also, including the RF climatic

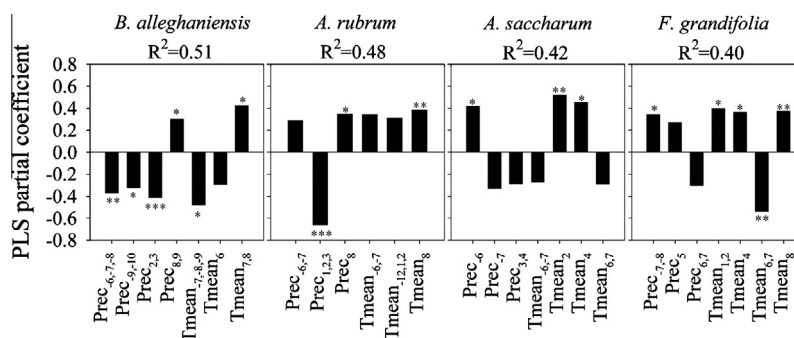


Fig. 4. Significant climatic variables in partial least squares (PLS) regression (one significant component) for the relationship between climatic variables and ring-width for each species. R^2 and significant variables (*) in the PLS regression are shown. Successive months having the same sign for their individual partial correlation coefficients were grouped (precipitation were summed while temperatures were averaged). Minus signs (–) indicate months of the preceding year, while & indicate months of the current year. Prec: monthly precipitation; Tmean: mean monthly temperatures. Months are indicated by numbers.

Table 2

Model comparison of Climate-implicit, Climate-explicit 30-year and Climate-explicit RF models for *B. alleghaniensis*, *A. rubrum*, *A. saccharum* and *F. grandifolia* based on AIC_c values, Akaike weights and evidence ratios. Only the best models (i.e., those with a $\Delta\text{AIC}_c < 2$) are shown for each species. R^2 between observed and predicted values are shown along with p -values. The mathematical form of each term is indicated.

Species	Type	Model	SE	VE	DE	BAE	R^2	AIC _c	ΔAIC_c	Akaike weight	Evidence ratio
<i>B. alleghaniensis</i>	Clim_exp_RF	23	Logi	None	Lin	Exp	0.36 [*]	9422.807	0.000	0.565	1.000
	Clim_exp_RF	19	Logi	Lin	Lin	Exp	0.36 [*]	9424.060	1.252	0.302	1.871
<i>A. rubrum</i>	Clim_imp	30	Lin	None	None	Exp	0.26 [*]	9219.845	0.000	0.044	1.000
	Clim_imp	46	Lin	None	None	Logn	0.26 [*]	9219.928	0.084	0.043	1.043
	Clim_imp	31	Quad	None	None	Exp	0.25 [*]	9220.173	0.329	0.038	1.179
	Clim_exp_30	47	Logi	None	None	Logn	0.26 [*]	9220.277	0.432	0.036	1.241
	Clim_imp	63	Quad	None	None	Logi	0.26 [*]	9220.399	0.554	0.034	1.319
	Clim_exp_30	63	Logi	None	None	Logi	0.26 [*]	9220.424	0.580	0.033	1.336
	Clim_imp	47	Quad	None	None	Logn	0.26 [*]	9220.484	0.639	0.032	1.376
	Clim_imp	62	Lin	None	None	Logi	0.24 [*]	9220.600	0.755	0.030	1.459
	Clim_exp_30	30	Quad	None	None	Exp	0.24 [*]	9220.719	0.874	0.029	1.548
	Clim_exp_30	46	Quad	None	None	Logn	0.24 [*]	9220.922	1.078	0.026	1.714
	Clim_exp_30	62	Quad	None	None	Logi	0.24 [*]	9221.079	1.235	0.024	1.854
	Clim_exp_30	31	Logi	None	None	Exp	0.26 [*]	9221.135	1.291	0.023	1.907
	Clim_exp_30	39	Logi	None	Lin	Logn	0.25 [*]	9221.403	1.559	0.020	2.180
	Clim_exp_30	55	Logi	None	Lin	Logi	0.27 [*]	9221.483	1.639	0.020	2.269
	Clim_imp	26	Lin	Lin	None	Exp	0.25 [*]	9221.496	1.651	0.019	2.283
	Clim_imp	58	Lin	Lin	None	Logi	0.24 [*]	9221.587	1.742	0.019	2.389
	Clim_imp	22	Lin	None	Lin	Exp	0.24 [*]	9221.607	1.762	0.018	2.414
	Clim_exp_RF	63	Quad	None	None	Logi	0.26 [*]	9221.761	1.917	0.017	2.607
<i>A. saccharum</i>	Clim_exp_RF	55	Logi	None	Lin	Logi	0.38 [*]	22894.122	0.000	0.656	1.000
<i>F. grandifolia</i>	Clim_exp_RF	58	Quad	Lin	None	Logi	0.42 ^{**}	2539.658	0.000	0.727	1.000

Abbreviations: SE: size effect; VE: vegetation effect; DE: drainage effect; BAE: basal area effect; Lin: linear; Quad: quadratic; Logi: logistic; Logn: lognormal, Clim_imp: Climate-implicit model; Clim_exp_30: Climate-explicit-30-years model; Clim_exp_RF: Climate-explicit-RF model.

* 0.05.

** 0.01.

*** 0.001.

Table 3

Sum of Akaike weights by model type (i.e., depending on how climate is accounted for, as well as whether size, vegetation type, drainage class and basal area are included). Bold characters indicate the highest values by species, within each set of comparisons.

Type	<i>B. alleghaniensis</i>	<i>A. rubrum</i>	<i>A. saccharum</i>	<i>F. grandifolia</i>
Climate-implicit	0.0000	0.5602	0.0000	0.0000
Climate-explicit-30-years	0.1187	0.3428	0.0000	0.0001
Climate-explicit-RF	0.8813	0.0970	1.0000	0.9999
With SE	0.9939	0.8595	1.0000	0.9835
Without SE	0.0061	0.1405	0.0000	0.0165
With VE	0.3501	0.2880	0.2189	0.9917
Without VE	0.6499	0.7120	0.7811	0.0083
With DE	0.9997	0.2816	0.9917	0.2460
Without DE	0.0003	0.7184	0.0083	0.7540
With BAE	1.0000	0.9553	1.0000	1.0000
Without BAE	0.0000	0.0447	0.0000	0.0000

Abbreviations: SE: size effect, VE: vegetation effect; DE: drainage effect; BAE: basal area effect.

variables did not significantly improve performance of models 39, 46, 47 and 63.

For *A. saccharum*, Climate-explicit-RF model 55 outperformed every tested model (Table 2). This model included logistic size and basal area effects as well as a linear drainage effect. Model comparison for *F. grandifolia* showed that Climate-explicit-RF model 58, which has a quadratic size effect, a linear vegetation effect and a logistic basal area effect, provided the best fit for the data.

For most models, preferred equation forms for the size effect and basal area effect were either logistic, quadratic or exponential. Linear and lognormal shapes were never retained as best models (Table 4). For *B. alleghaniensis*, Akaike weights showed that good models had a logistic size effect and an exponential basal area effect. Mixed results were obtained for *A. rubrum*: most support was observed for quadratic size effect and logistic basal area effect but some support was also observed for linear size effect, and

exponential or lognormal basal area effect. Such results are in agreement with our observation that best models comprised a mix of Climate-explicit-30-years, Climate-implicit and Climate-explicit-RF models. Both logistic size effect and basal area effect had support to model *A. saccharum* growth. Similar results were obtained for *F. grandifolia* but with a quadratic size effect term.

For *B. alleghaniensis*, *A. saccharum* and *F. grandifolia*, Climate-implicit model showed less bias than Climate-explicit-30-years and Climate-explicit-RF models (Table 5). For *A. rubrum*, the lowest bias was observed for the Climate-explicit-RF model. Our result showed that most Climate-explicit-RF models had smaller values of AIC_c but always had a larger bias. AIC_c value identifies the model that best compromises between minimizing bias and improving accuracy (Akaike, 1974). Thus, it is likely that a model identified as “best” had a higher bias than the second model if the variance is lower than the second. Finally, Climate-implicit models always

Table 4

Sum of Akaike weights of models with a size effect, and with a basal area effect, as a function of the mathematical form of the term. Bold characters are used to highlight the highest value by species and by type of comparison.

Variable	Shape	<i>B. alleghaniensis</i>	<i>A. rubrum</i>	<i>A. saccharum</i>	<i>F. grandifolia</i>
SE	Linear	0.1221	0.2868	0.0000	0.0108
	Logistic	0.8705	0.1847	0.9948	0.0055
	Quadratic	0.0012	0.3880	0.0052	0.9672
BAE	Linear	0.0000	0.0254	0.0000	0.0000
	Exponential	0.8824	0.3021	0.0004	0.0124
	Logistic	0.0600	0.3262	0.8412	0.9778
	Lognormal	0.0576	0.3015	0.1585	0.0097

underestimated tree growth (based on negative bias values for all four species).

4. Discussion

In our study, the inclusion 30-year mean annual temperature and precipitation in tree growth models did not provide a better or comparable fit to data as when climate was represented by current and previous year monthly climatic variables. Such a finding is in agreement with the hypothesis that annual ring growth is more closely coupled to previous and current year climatic conditions, i.e. those that prevailed during ring formation, than to long term average values.

Significant associations between radial growth and monthly climate of the current and previous years were also found by other authors for species such as *A. saccharum* (Tardif et al., 2001), *Tsuga mertensiana* (Gedalof and Smith, 2001), *Picea mariana* (Hofgaard et al., 1999; Girard et al., 2011a), *Pinus halepensis* (Girard et al., 2011b; Girard et al., 2012) and *P. tremuloides* (Huang et al., 2010). There are very few papers that used response function results or current- and previous year monthly data as climatic input to model tree growth. It is likely due to the availability of the data and the ease at which users of growth models can obtain such data. Another hurdle for climate-driven growth models is that the empirically based relationships between tree growth and climate are limited to the range in variability of the historical climatic conditions.

A large amount of diameter growth is predetermined in the cambium and the climatic conditions prevailing during the year prior to ring formation may have significant impacts on achieved diameter growth (Schweingruber, 1996; Briffa et al., 2001). Accordingly, response function for the species tested in this study, showed some significant associations with precipitation and/or temperature values of previous summer and/or fall. For example, increased amount of precipitation during the previous year can act as nutrient pool for the trees during the current year, promoting growth. The most consistent relationship identified by the response function was the tendency for higher growth to be associated with higher temperatures and abundant precipitation during April and August of the current year. This relationship reflects the importance of air temperature during the beginning of the growing season in regulating soil temperatures, rates of respiration and photosynthesis, metabolic processes, and carbohydrate production (Bonan and Sirois, 1992; Hofgaard et al., 1999; Kramer et al., 2000;

Tardif et al., 2001; Lisi et al., 2008). The association between low precipitation during summer and large ring-width has already been documented in literature (Rathgeber et al., 2000; Deslauriers and Morin, 2005). Most of the time, low summer precipitation are linked with high temperatures and insolation which may promote tree growth. A higher temperature during the month of August is linked with late-growth and lignification processes in eastern Canada. Higher temperature in late summer provides better conditions for late-wood formation, cell elongation and lignification (Deslauriers et al., 2003; Deslauriers and Morin, 2005).

Known physiological relationships between ring growth and recent climate conditions might explain why Climate-explicit-RF outperformed Climate-explicit-30-years models in most species in this study. A 30-year average of annual climatic conditions is likely to include a lot of insignificant variables (i.e., irrelevant climatic information) that have no ecophysiological effect on tree growth, as well as results for all the years before the year prior to ring formation. The inclusion of insignificant months in the model only adds noise and decreases accuracy when predicting tree growth. Not only are some months less relevant to observed ring growth, but so are the many years of data that are included in a 30-years average. We suggest that the 30 year mean annual climatic variables should be used to regionalize tree growth predictions from growth models rather than to make projections in a climate change context.

Comparing multiple equation forms for the size and competition terms allowed us to determine the form that provided the best fit for each model. Five different equation forms were considered for the competition effect because of their simplicity for parameterization in growth models. For most species, a logistic function provided a good fit for the size and basal area effects. The good performance of a logistic equation form to describe ecophysiological processes had been reported by Morris and Silk (1992). The quadratic form used to model the size effect also provided a good fit to data for *A. rubrum* and *F. grandifolia*.

Inclusion of a linear drainage effect in models significantly improved the fit for *B. alleghaniensis* and *A. saccharum* but was never retained for *F. grandifolia*. The rare occurrence of *F. grandifolia* in the mixed stands used to calibrate the growth models may be responsible for the poor fit with DE. Also, yearly water needs of *F. grandifolia* are lower than those for *B. alleghaniensis* and *A. saccharum*, very likely explaining why drainage was not retained in the growth models (Erdmann, 1990; Godman et al., 1990; Tubbs and Houston, 1990).

Table 5

Bias and accuracy of the best models for each species. For *B. alleghaniensis*, results from model 55 are shown for Climate-implicit and Climate-explicit 30-year models whereas model 23 is shown for Climate-explicit model (For *A. rubrum*, *A. saccharum* and *F. grandifolia* results from model 63, 55 and 58 are shown respectively for all three families of models).

Type	<i>B. alleghaniensis</i>			<i>A. rubrum</i>			<i>A. saccharum</i>			<i>F. grandifolia</i>		
	Model	Bias	Accuracy	Model	Bias	Accuracy	Model	Bias	Accuracy	Model	Bias	Accuracy
Climate-implicit	55	−0.128	3.687	63	−0.174	4.945	55	−0.013	19.204	58	−0.128	3.687
Climate-explicit-30-years	55	−0.393	11.315	63	−3.808	108.048	55	0.371	16.957	58	−4.746	68.123
Climate-explicit-RF	23	−0.465	13.376	63	−0.162	4.607	55	0.198	9.077	58	−0.516	7.400

4.1. Management implications

Despite the fact that climate change is expected to have significant impacts on forest ecosystems over the next 100 years (Solomon et al., 2007), tree growth projections and management decisions made today are still often based on the implicit assumption that the climate will remain relatively stable through time. Forest management is increasingly being viewed as a central component of strategies for climate change mitigation and adaptation (Solomon et al., 2007; D'Amato et al., 2011). Growth model developers did not assume that future climatic effects on growth were negligible. Rather, past growth as influenced by past climatic conditions, could be extrapolated beyond the calibration period under the assumption that future climatic conditions would be similar to past climatic conditions. The use of dendroclimatological results to modify tree potential growth in a traditional growth model is the first step towards a new lineage of tree growth models.

However, forest managers should use those models in the right context. It is unlikely that Climate-explicit-RF growth models may be useful without adequate calibration for a particular study area. Adequate cross-validation of the response function over the years is also necessary to evaluate the stability of the relationship between climate and tree-ring growth (Hughes et al., 2010). Our Climate-explicit-RF models outperformed the two other types of growth models for modeling past and current radial growth. Therefore, projections for the future should be possible but further validation of the method is necessary before widespread use for forest growth projections.

The traditional approach for dendroclimatic reconstructions is to calibrate tree rings with climate variables using some type of linear regression technique (Cook and Kairiukstis, 1990). However, tree growth response to climate is often nonlinear (Fritts, 2001). A few recent studies demonstrated that the response of tree growth to climate is likely non-linear in a changing climate regime.) The findings imply that (1) monthly climatic conditions should be incorporated into growth models under a rapidly changing climate regime, and (2) to successfully incorporate these effects requires models which are flexible given the non-linear response of tree growth to changing climatic conditions. With growing concern regarding potential climate change, the most useful models will be sensitive to key effects of climate change on tree and stand development over long time periods. This will be fundamental to addressing questions of sustainability of forest management.

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Appendix A. Supplementary materials

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.foreco.2014.03.014>.

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